

4 Results

4.1 Fact 1: Employment for young workers has declined in AI-exposed occupations

Consider software engineers and customer service agents, two occupations frequently considered to be highly exposed to generative AI tools. Media attention has raised the specter of widespread employment disruption for young software engineers (Thompson, 2025; Raman, 2025; Allen, 2025; Horowitch, 2025).

Figure 1 shows employment changes by age group for these occupations, normalized to 1 in October 2022. While the types of work and workers in each of these occupations differs in many ways, both occupations present a similar pattern: employment for the youngest workers declines considerably after 2022, while employment for other age groups continues to grow. By September 2025, employment for software developers aged 22-25 declined nearly 20% compared to its peak in late 2022. Figure A1 shows that a similar pattern holds for computer occupations and service clerks more generally.

Figure A2 shows four other professions as case studies, spanning varying levels of AI exposure according to Eloundou et al. (2024). Marketing and sales managers, in the fourth quintile of AI exposure, show a decline in employment for young workers much like the case of software and customer service, albeit with smaller magnitudes. Front-line production and operations supervisors, in quintile 3, show an increase in employment for young workers, though the growth in employment is smaller than the increase for workers over age 35.

In contrast, the trends for occupations that Eloundou et al. (2024) rated as less exposed do not fit the pattern of the more exposed occupations. Stock clerks and order fillers, in quintile 2, show no obvious difference by age. Strikingly, the series for health aides, comprising nursing aides, psychiatric aides, and home health aides, show a quite different trend from software or customer service: employment for young workers has been growing *faster* than for older workers.

Figure 2 shows these patterns hold more generally across professions. The top left plot shows a divergence in employment outcomes for more and less exposed occupations for workers aged 22-25, with more exposed occupations experiencing declining employment. For older age groups, we find

much less marked differences in employment growth across AI exposure quintiles.¹⁵

4.2 Fact 2: Though overall employment continues to grow, employment growth for young workers has been stagnant

Overall employment in the ADP data remains robust, coinciding with a low national unemployment rate in the post-pandemic period. However, Figure A4 suggests some leveling off in employment growth for young workers relative to other age groups, consistent with recent discussion of a worsening job market for entry-level workers (Chen, 2025; Federal Reserve Bank of New York, 2025).

Figure A5 offers insight into how these trends relate to AI exposure. For each age group, employment growth from late 2022 to September 2025 was 5-13% for the lowest three AI exposure quintiles, with no clear ordering in employment growth by age. In contrast, for the highest two exposure quintiles employment for 22-25 year olds declined by 6% between late 2022 and September 2025, while employment for workers aged 35-49 grew by over 8%. These results show that declining employment in AI-exposed jobs is driving tepid overall employment growth for workers between the ages of 22 and 25.

While these findings suggest divergent employment outcomes by AI exposure for young workers, we caution the trends observed in these first two facts could be driven by other changes in the US economy. Our subsequent facts evaluate the robustness of the results to alternative analyses.

4.3 Fact 3: Entry-level employment has declined in applications of AI that *automate* work, with muted changes for *augmentation*

AI exposure can either complement or substitute for labor. These may have very different implications for the labor market (Brynjolfsson, 2022).

To assess how employment patterns differ based on the complementarity or substitutability of AI with labor, we use data on generative AI usage from the Anthropic Economic Index (Handa et al., 2025). The Index provides an estimate of the share of queries that pertain to each occupation. In addition, for each task it reports estimates of the share of queries pertaining to that task that are “automative,” “augmentative,” or none of the above. We use these classifications as estimates

¹⁵Figure A3 shows that declining employment for young workers spans a range of occupations. Close to 70% of occupations in the first exposure quintile see rising early career employment between October 2022 and September 2025, compared to less than half of occupations in the fifth quintile.

of whether the usage of AI in an occupation is primarily a substitute or complement for labor. Table A1 shows example occupations that are in the highest and lowest exposure category for each measure.

Panel A of Figure 3 shows employment changes by overall prevalence of related Claude queries for 22-25 year-olds.¹⁶ The patterns match the findings using the Eloundou et al. (2024) measures closely. Panel B likewise shows that the occupations with the highest estimated automation shares have experienced declining employment for the youngest workers.

In contrast, Panel C indicates that the occupations with the highest estimated augmentation shares have *not* experienced a similar pattern. Employment changes for young workers are not ordered by augmentation exposure, as the fifth quintile has among the fastest employment growth. The findings are consistent with automative uses of AI substituting for labor while augmentative uses do not.¹⁷

4.4 Fact 4: Employment declines for young, AI-exposed workers remain after conditioning on firm-time shocks

While our results so far are consistent with the hypothesis that generative AI is causing a decline in entry-level employment, there are plausible alternative explanations. One class of explanations is that our patterns are explained by industry- or firm-level shocks correlated with sorting patterns by age and measured AI exposure. For example, one possibility is that young workers with high measured AI exposure are disproportionately likely to sort to firms heavily susceptible to interest rate increases.

We test for a class of such confounders by controlling for a rich set of fixed effects. For each age group, we estimate the Poisson regression

$$\log(E[y_{f,q,t}]) = \sum_{q' \neq 1} \sum_{j \neq -1} \gamma_{q',j} 1\{t = j\} 1\{q' = q\} + \alpha_{f,q} + \beta_{f,t} + \epsilon_{f,q,t} \quad (4.1)$$

¹⁶See Figures A6 through A8 for other age groups.

¹⁷Occupations in the first two augmentation quintiles have very low Claude usage (0.01% and 0.09% of conversations for the average occupation, respectively), with a high share of conversations classified as neither automative nor augmentative. In contrast, occupations in the third through fifth quintiles average 0.47%, 0.39%, and 0.33% of Claude conversations. For the automation measure, overall Claude usage increases on average with the automation share, with the lowest exposure group averaging 0.05% of conversations and the highest group averaging 0.73%. Figures A9 and A10 show that automation and augmentation results are similar when dropping occupations with low overall Claude usage.

f indexes firms, q indexes Eloundou et al. (2024) exposure quintiles, and t indexes months, with $t = -1$ corresponding to October 2022. The outcome variable $y_{f,q,t}$ is employment in f, q, t . Equation 4.1 is a Poisson event study regression controlling for firm-quintile effects, $\alpha_{f,q}$, and firm-time effects, $\beta_{f,t}$. The firm-time effects absorb aggregate firm shocks that impact each exposure quintile equally. The firm-quintile effects adjust for baseline differences in hiring across quintiles within the firm. The coefficients of interest, $\gamma_{q,t}$, measure differential changes in employment growth across quintiles after accounting for firm-time effects and firm-quintile effects.¹⁸

We run this regression separately for each age group. For each regression, we restrict to firms that hire at least 10 workers within the age group in every period of the sample. Further, $\sum_t y_{f,q,t}$ must equal at least 100 for each q , meaning that the firm must at least employ on average about 2 workers from each exposure quintile across months in the sample.¹⁹ Standard errors are clustered by firm.

Results are in Figure 4, which plots the $\gamma_{q,t}$ coefficients for each age group. For workers aged 22-25, estimates for higher quintiles are large and statistically significant, with a 15 log point decline in relative employment comparable in magnitude to the estimates in the raw data in Figure 2. Estimates for other age groups are generally much smaller in magnitude and not statistically significant. These findings imply that the employment trends we observe are not driven by differential shocks to firms that employ a disproportionate share of AI-exposed young workers.

One alternative confounder that would not be controlled for with firm-time effects is that even conditional on the firm workers with high AI exposure were excessively hired after the COVID-19 pandemic, leading to a subsequent contraction in their hiring. To assess such alternatives we consider various other robustness checks in Section 4.6, such as removing computer occupations and conditioning on whether the occupation is amenable to work from home.

¹⁸Because of zero counts in the outcome variable, we estimate a Poisson regression instead of an OLS regression in logs following guidance from Chen and Roth (2024).

¹⁹Results are not sensitive to these restrictions, though there must be at least one non-zero value in each firm-month and each firm-quintile for observations to not get dropped in the Poisson regression.